

Tests of numerical errors between Matlab and Python functions

This document is a report of all the numerical errors that were measured between Numpy/Scipy and Matlab's functions for sound processing .

First : Interpolations :

The Outer Middle Model contains two steps of interpolations , the two interpolated lists are the added in tfOuterMiddle, here is the error resulting of the same computation between Matlab and Numpy's interpolation.

1 : Outer Middle Model :

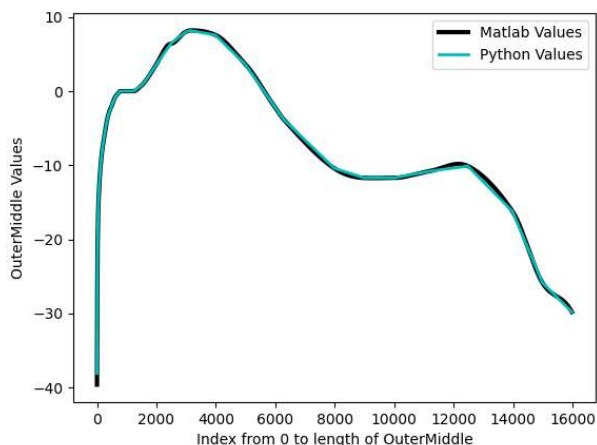
Here are the measures done on the list called OuterMiddle of the data_glasberg matlab file reproduced in python.

Maximum error for outerMiddle is : 1.5200000000000031

* Maximum error is $\max(\text{abs}(\text{error}))$

Average error for outerMiddle is 0.1826141705261848

Quadratic error for outerMiddle is : 0.06420167375087449



FILTERS :

Fir2 filter :

The parameters are 1997 model with Free model .

Maximum error for Filter is : 0.3200670631705158

Average error for Filter is 0.0009037093750747852

Quadratic error for Filter is : 0.00014586153447515663

